

T1K vs. Immuannot on the giraffe/whatshap 1000G pangenome-typing pipeline

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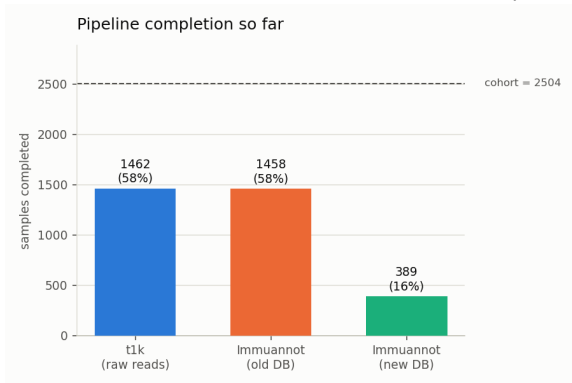
BioHackathon Japan 2026, HLA/pangenome group

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Four HLA typing runs on the same 1000G short-read cohort, compared pairwise

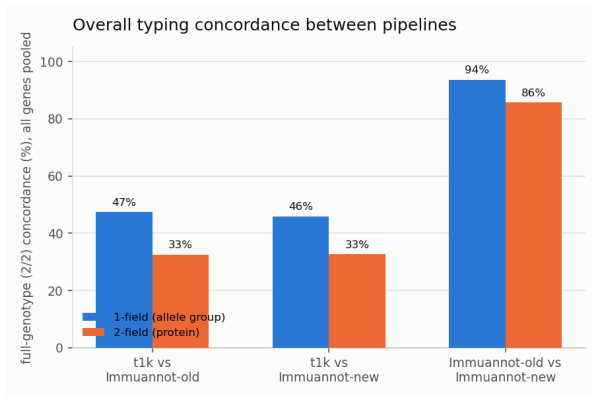
- ▶ **t1k**: run-t1k directly on raw FASTQs extracted per sample from the 1000G high-coverage CRAMs (mhc_giraffe/t1k); independent of the graph/phasing steps below
- ▶ **Immuannot, old DB**: vg giraffe alignment to leechuck's whole-MHC Minigraph-Cactus graph → vg call → WhatsHap read-backed phasing → bcftools consensus hap1/hap2 FASTAs → Immuannot against leechuck's frozen IPD-IMGT/HLA 3.55.0
- ▶ **Immuannot, new DB**: identical hap1/hap2 consensus FASTAs, re-annotated against a freshly built IPD-IMGT/HLA 3.65.0 (same release T1K uses) – isolates database-version effects only
- ▶ **leechuck_t1k**: leechuck's own run-t1k on raw 1000G reads, entirely outside this pipeline; 412 samples × 5 genes (A/B/C/DQB1/DRB1), merged against published truth genotypes
- ▶ Comparison: unordered allele pairs, truncated to 1-field (allele group) and 2-field (protein) resolution; Immuannot's :new suffix (no exact database match) stripped before truncation, not treated as a literal field

t1k and Immuannot each cover more than half of the 2,504-sample cohort so far



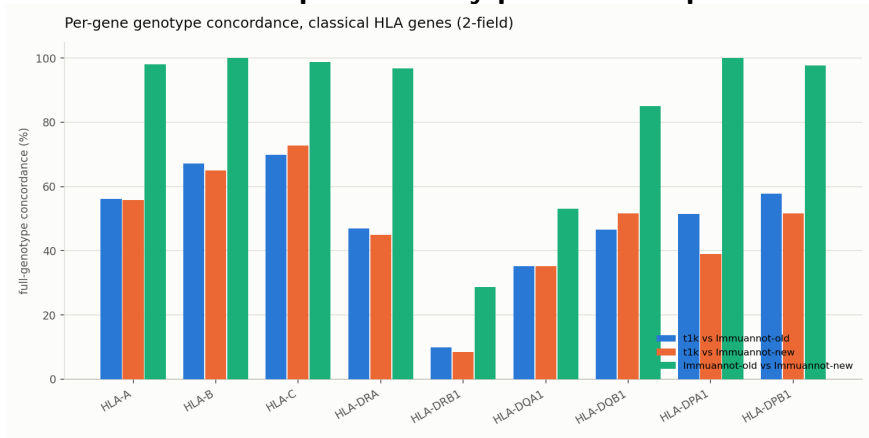
- ▶ t1k and Immuannot-old-DB run independently per sample and finish at the same pace (both $\approx 58\%$); Immuannot-new-DB is a smaller re-run launched later, still catching up (16%)
- ▶ All figures on later slides use whatever fraction of the cohort had finished at the time of analysis

Immuannot's old and new databases agree almost perfectly; t1k agrees with Immuannot only 1 call in 3



- ▶ Immuannot-old vs Immuannot-new: 86% full-genotype (2-field), 94% at 1-field – expected, since it is the same consensus sequence, only the reference database changed
- ▶ t1k vs Immuannot (either DB): only $\approx 32\%$ full-genotype concordance – Immuannot types off a phased short-read consensus (subject to phasing/assembly error), t1k types straight from reads

HLA-DRB1 concordance collapses in every pairwise comparison

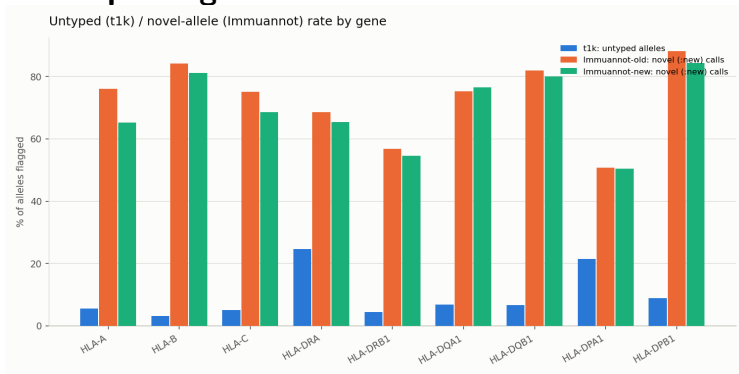


- ▶ A/B/C/DPA1/DPB1: Immuannot-old vs Immuannot-new $\geq 97\%$; t1k vs either Immuannot 50–74%
- ▶ HLA-DRB1: only 9–29% concordance for every pair, including Immuannot-old vs Immuannot-new (same consensus sequence!) – a paralog-confusable, hard-to-phase locus, not a database-version or typer artifact

How: genotype comparison at 1- and 2-field resolution

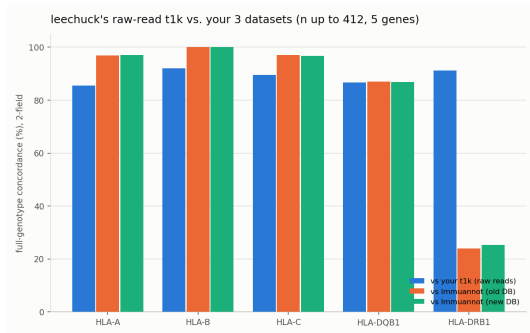
- ▶ Truncate each allele string to its first ($k=1$) or first two ($k=2$) colon-separated fields after the gene, e.g. HLA-A*29:02:01:01 $\rightarrow$ HLA-A*29:02
- ▶ Immuannot's :new suffix marks the point past which no exact database allele matched – HLA-A*01:new is only *1-field*-resolved, not a literal second field of “new”; stripped before truncation, and a 2-field comparison is skipped (not counted as a mismatch) when fewer than 2 real fields remain
- ▶ Genotype match: best of the two possible allele-pairings (hap1/hap2 vs allele1/allele2 order is arbitrary between pipelines); full match requires 2/2 alleles to agree
- ▶ Script: analysis/gc_typing_concordance.py; tables in results/tables/gc_*.tsv

65–88% of Immuannot's classical-gene calls are novel (:new), consistent with imperfect short-read phasing



- ▶ t1k rarely leaves a classical gene untyped (3–9%, DPA1/DRA higher at 21–25% – both have many database alleles close in sequence)
- ▶ Immuannot flags a novel allele on 65–88% of A/B/C/DQB1/DPB1 calls, old and new DB alike – the consensus sequence itself, not the database, is the limiting factor (matches the pipeline's own documented caveat: WhatsHap phasing can leave chimeric/ambiguous consensus for highly polymorphic genes)

leechuck's raw-read t1k agrees with Immuannot far better than this pipeline's own t1k does



- ▶ Same 5 genes, up to 412 samples (different, smaller subset than the whole-cohort figures above)
- ▶ vs. this pipeline's t1k: 85–92% – the two t1k runs mostly agree with each other
- ▶ vs. Immuannot (either DB): 87–100% for A/B/C/DQB1 – markedly higher than this pipeline's own t1k-vs-Immuannot ($\approx$ 54–70% on these genes, previous whole-cohort figures); DRB1 still collapses (24–25%)
- ▶ Read: when the two t1k runs disagree, it is usually this pipeline's t1k that is the outlier relative to Immuannot – investigated on the next slide

Digging in: same t1k binary, same database, different read extraction

- ▶ Both t1k runs use the identical database: diff of the two `hlaidx_dna_seq.fa` files (29,429 alleles each, IPD-IMGT/HLA 3.65.0) is non-empty only because records are in a different order – sorted, byte-identical. Rules out a database-version explanation
- ▶ `mhc_pipeline.sbatch`'s t1k step even reuses leechuck's own conda env (`mm/envs/typing/bin`) – same t1k version too
- ▶ Where they differ: **read extraction**. leechuck's `kg_mhc_reads.sbatch` pulls reads aligned to the MHC region *and* the 16 GRCh38 `chr6*_alt` contigs *and* the 525 HLA-* allele contigs (542 regions total) straight from the 1000G high-coverage CRAMs
- ▶ This pipeline's extraction (`03_subsample_1000g_cram.sh`) pulls the MHC region and the 525 HLA-* contigs, but *not* the 16 `chr6*_alt` contigs (it adds unmapped reads instead, which leechuck's does not)

Root cause: this pipeline's read extraction misses the chr6 MHC ALT haplotype contigs

- ▶ Checked directly on HG00097 (one of the discordant samples, homozygous in leechuck's t1k, one allele untyped in ours): **224,460 reads** in its 1000G CRAM align to the 16 chr6*_alt contigs alone
- ▶ Those reads never reach this pipeline's t1k input at all – for a sample whose true haplotype is closer to an ALT scaffold than to the GRCh38 primary path, this can substantially reduce effective read support for the divergent allele
- ▶ Consistent with the observed failure mode: t1k calls one allele confidently and either repeats it (leechuck, full read set) or leaves the second slot untyped (ours, reduced read set) – an abundance/coverage effect, not a wrong call
- ▶ Fix: add the 16 chr6*_alt contigs to hla_decoy_contigs.txt / the extraction region list and re-run t1k; cheap, since it only touches the read-extraction step

Take-home

- ▶ Immuannot's database version (3.55 vs 3.65) barely matters here (86–94% agreement); the consensus sequence quality from graph alignment + WhatsHap phasing is the real bottleneck, especially for A/B/C/DQB1/DPB1 (65–88% flagged novel) and DRB1 (concordance collapses even DB-vs-DB)
- ▶ t1k-vs-Immuannot discordance ($\approx 32\%$ whole-cohort) is expected and matches the pipeline's own documented caveat about imperfect phasing – t1k is the more trustworthy call for benchmarking, per that same caveat
- ▶ A second, independent t1k run (leechuck's) agrees with Immuannot noticeably better than this pipeline's t1k does; traced to a read-extraction gap (missing chr6 MHC ALT contigs), not a tool, database, or pangenome-pipeline difference – fixable
- ▶ Open: re-run this pipeline's read extraction with the ALT contigs included and re-check t1k-vs-leechuck_t1k concordance; revisit DRB1 specifically once Immuannot-new-DB finishes the full cohort
- ▶ Script, figures, tables: `pangenome-bh26/hla/analysis/gc_typing_concordance.py`, `results/figures/gc_*.png`, `results/tables/gc_*.tsv`